

NORTH CENTRAL WEST VIRGINIA, WV, USA

WMO: 724175

Lat: 39.296N

Lon: 80.229W

Elev: 1203

StdP: 14.07

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 03802

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.0	11.8	-2.7	5.0	10.0	1.3	6.1	14.7	23.2	43.8	20.4	40.7	4.9	270	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	89.3	72.6	87.0	71.7	84.6	70.3	75.4	84.5	74.2	82.7	73.0	81.0	7.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	125.9	80.1	71.8	122.7	79.2	70.3	116.7	77.6	39.6	85.1	38.4	82.4	37.2	81.1	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.0	16.8	14.7	DB	-0.6	91.8	5.2	3.0	-4.3	94.0	-7.3	95.7	-10.2	97.4	-14.0	99.5	
			WB	-1.3	77.6	5.1	1.6	-4.9	78.8	-7.9	79.7	-10.8	80.6	-14.5	81.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	31.6	34.6	43.0	54.8	63.0	70.2	73.3	72.4	66.5	55.1	44.8	37.0
	DBStd	17.11	12.58	11.23	10.69	9.37	7.64	5.53	4.68	4.53	6.92	8.81	9.14	10.48
	HDD50	2038	584	444	273	57	5	0	0	0	0	48	206	421
	HDD65	4911	1034	851	683	322	132	19	3	64	326	606	868	
	CDD50	3486	14	14	56	201	409	605	723	694	496	205	51	18
	CDD65	883	0	0	1	17	70	174	260	232	110	19	0	0
	CDH74	7246	0	2	17	216	666	1421	2024	1766	951	177	6	0
	CDH80	2091	0	0	1	34	153	410	640	529	286	38	0	0
Wind	WSAvg	5.4	7.0	7.0	6.8	6.7	5.1	4.6	4.1	3.6	3.8	4.6	5.5	6.1
Precipitation	PrecAvg	46.9	3.9	3.1	4.0	3.7	4.9	4.7	5.3	3.7	3.7	2.9	3.3	3.7
	PrecMax	64.8	6.7	5.3	7.6	6.1	10.2	10.5	11.3	6.7	7.1	5.0	6.9	8.5
	PrecMin	39.3	1.7	1.0	1.5	1.9	1.9	2.0	2.8	1.9	0.9	1.1	0.9	1.7
	PrecStd	6.7	1.4	1.3	1.6	1.1	2.0	2.0	2.2	1.3	1.7	1.3	1.3	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.8	70.5	76.6	83.9	87.2	90.5	91.4	90.9	90.6	84.7	73.5	67.6
		MCWB	56.6	57.1	59.8	62.0	69.4	70.9	74.9	73.2	69.8	68.1	60.5	58.5
	2%	DB	61.6	63.2	71.5	80.0	83.9	87.4	88.6	88.2	86.6	78.8	69.3	63.0
		MCWB	54.0	53.0	56.6	61.0	67.2	71.2	73.9	72.7	69.9	64.9	57.1	54.3
	5%	DB	56.6	57.8	66.0	75.7	81.3	84.3	86.2	85.5	83.3	74.4	64.2	58.2
		MCWB	51.1	50.3	53.7	59.2	66.6	70.4	72.4	71.4	68.3	62.8	54.3	52.2
	10%	DB	51.2	53.3	61.0	71.7	78.0	82.0	83.7	82.7	80.1	70.6	60.9	54.1
		MCWB	46.2	46.7	51.5	57.2	65.0	69.4	71.3	70.5	67.0	60.9	53.1	49.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	59.6	61.5	66.2	72.7	75.3	77.6	77.0	74.6	71.3	62.8	59.9
		MCDB	64.7	66.7	74.0	78.3	83.3	84.1	87.7	86.5	85.0	79.5	71.4	65.8
	2%	WB	55.3	55.5	58.8	63.6	70.2	73.6	75.8	75.1	72.4	68.0	59.4	56.7
		MCDB	60.2	61.3	67.7	74.0	80.4	82.5	85.3	83.7	81.5	76.1	66.4	61.2
	5%	WB	51.9	51.1	56.2	61.5	68.4	72.2	74.5	73.7	70.7	65.0	56.7	53.3
		MCDB	56.0	56.6	63.5	71.5	77.6	81.4	82.9	81.8	78.6	71.9	62.6	57.2
	10%	WB	46.3	46.4	53.0	59.5	66.6	70.9	73.2	72.4	69.1	62.0	53.7	49.3
		MCDB	50.3	51.9	59.8	69.1	74.9	79.7	80.5	79.8	76.2	68.5	59.6	53.7
Mean Daily Temperature Range	5% DB	MDBR	15.9	17.0	19.4	22.2	21.2	19.8	18.7	19.5	21.3	20.6	19.1	15.6
		MCDBR	21.0	24.9	27.1	28.3	25.7	22.9	21.5	22.8	25.7	25.9	26.0	22.9
		MCWBR	16.0	18.3	16.2	14.1	12.0	10.0	8.9	9.3	10.9	13.7	16.4	16.6
	5% WB	MCDBR	19.8	22.2	22.7	22.6	21.3	19.4	17.9	18.3	20.3	21.3	20.7	20.7
		MCWBR	16.1	18.1	15.4	12.8	10.7	9.4	8.5	8.7	9.5	12.1	14.9	16.9
Clear-Sky Solar Irradiance	taub	0.301	0.320	0.346	0.385	0.429	0.461	0.482	0.497	0.407	0.363	0.324	0.300	
	taud	2.469	2.403	2.403	2.322	2.205	2.152	2.111	2.039	2.285	2.408	2.492	2.510	
	Ebn at Noon	277	286	289	283	271	261	254	246	265	266	266	270	
	Edh at Noon	25	31	34	39	45	48	49	52	38	30	24	23	
All-Sky Solar Radiation	RadAvg	526	781	1116	1481	1680	1853	1788	1658	1377	967	670	462	
	RadStd	61	118	89	126	141	133	106	93	150	115	82	52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.17	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+196	N/A

Nomenclature: See separate page